# Spotify Tracks Dataset

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
print(df1.columns)
```

```
Index(['track_id', 'artists', 'album_name', 'track_name', 'popularity',
       'duration_ms', 'explicit', 'danceability', 'energy', 'key', 'loudness',
       'mode', 'speechiness', 'acousticness', 'instrumentalness', 'liveness',
       'valence', 'tempo', 'time_signature', 'track_genre'],
      dtype='object')
```

1. Which genre has the most songs?

```python
#Which genre has the most songs?
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
grouped=df1.groupby("track_genre")
res=grouped.value_counts()
print(res.head(1))
```

```
track_genre  track_id                artists          album_name     track_name     popularity  duration_ms  explicit  danceability  energy
acoustic     00DuOxungwHwVUX2m37P0P  Chord Overstreet What You Need  What You Need  53          181786       False     0.519         0.264
Name: count, dtype: int64
```

2. Which genre is most popular?

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
genre_popularity = df.groupby("track_genre")["popularity"].mean()

most_popular_genre = genre_popularity.idxmax()

print("Most Popular Genre:", most_popular_genre)
```

```
Most Popular Genre: pop-film
```

3. Which artist has the most songs?

```
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
top_artist = df1["artists"].value_counts().idxmax()
song_count = df1["artists"].value_counts().max()

print("Artist:", top_artist)
print("Number of Songs:", song_count)
```

```
Artist: The Beatles
Number of Songs: 279
```

4. Which artist has the highest average popularity?

```
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
grouped=df1.groupby("artists")["popularity"].mean()
top_artist =grouped.idxmax()
avg_popularity = grouped.max()

print("Artist:", top_artist)
print("Average Popularity:", avg_popularity)
```

```
Artist: Sam Smith;Kim Petras
Average Popularity: 100.0
```

5. Are danceable songs more popular?

```
import pandas as pd
df=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
corr = df["danceability"].corr(df["popularity"])

print("Correlation between Danceability and Popularity:", corr)
if corr > 0:
    print("Danceable songs tend to be more popular.")
elif corr < 0:
    print("Danceable songs tend to be less popular.")
else:
    print("No relationship between danceability and popularity.")
```

```
Correlation between Danceability and Popularity: 0.035448134751385973
Danceable songs tend to be more popular.
```

6. Are energetic songs more popular?

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
corr=df1["energy"].corr(df1["popularity"])
print("Correlation between energy and Popularity:", corr)
if corr > 0:
    print("energetic songs tend to be more popular.")
elif corr < 0:
    print("energetic songs tend to be less popular.")
else:
    print("No relationship between energetic and popularity.")
```

```
Correlation between energy and Popularity: 0.0010561362996985154
energetic songs tend to be more popular.
```

7. What percentage of songs are explicit?

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
explicit_count=df1["explicit"].sum()
total_count=len(df1)
percntage=(explicit_count/total_count)*100
print("the percentage of song explicit is:",percntage)
```

```
the percentage of song explicit is: 8.55
```

8. Which genre contains the happiest songs?

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
grouped=df1.groupby("track_genre")["valence"].mean()
print(grouped.sort_values(ascending=False).head(1))
```

```
track_genre
salsa    0.814838
Name: valence, dtype: float64
```

9. Which genre contains the saddest songs?

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
grouped=df1.groupby("track_genre")["valence"].mean()
print(grouped.sort_values(ascending=True).head(1))
```

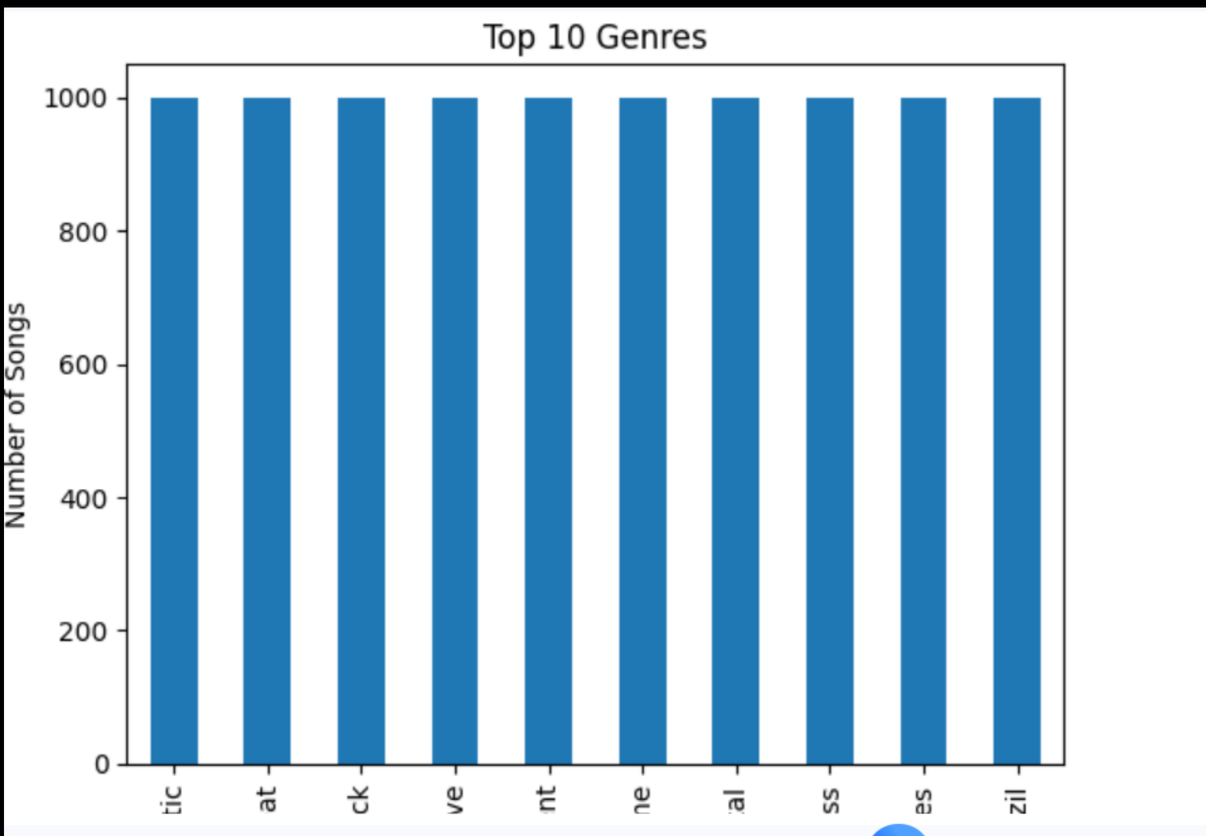
```
track_genre
sleep    0.058188
Name: valence, dtype: float64
```

10. What are the top 20 songs in the dataset?

```python
import pandas as pd
df1=pd.read_csv("/content/spotify-tracks-dataset-detailed.csv")
grouped=df1.groupby("track_name")["popularity"].max()
print(grouped.sort_values(ascending=False).head(20))
```

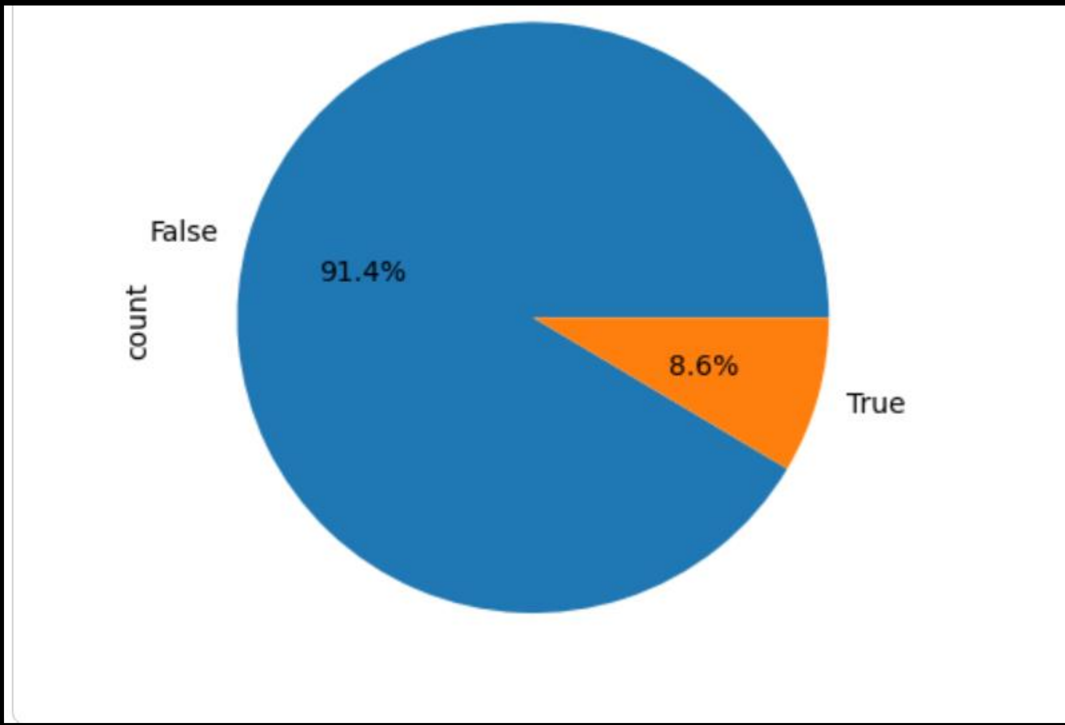
```
track_name
Unholy (feat. Kim Petras)                 100
Quevedo: Bzrp Music Sessions, Vol. 52      99
I'm Good (Blue)                            98
La Bachata                                 98
Tití Me Preguntó                           97
Me Porto Bonito                            97
Efecto                                     96
Under The Influence                        96
I Ain't Worried                            96
As It Was                                  95
Ojitos Lindos                              95
Glimpse of Us                              94
Moscow Mule                                94
PROVENZA                                   93
Sweater Weather                            93
Neverita                                   93
```

```python
import matplotlib.pyplot as plt

top_genres = df["track_genre"].value_counts().head(10)

top_genres.plot(kind="bar")

plt.title("Top 10 Genres")
plt.xlabel("Genre")
plt.ylabel("Number of Songs")

plt.show()
```



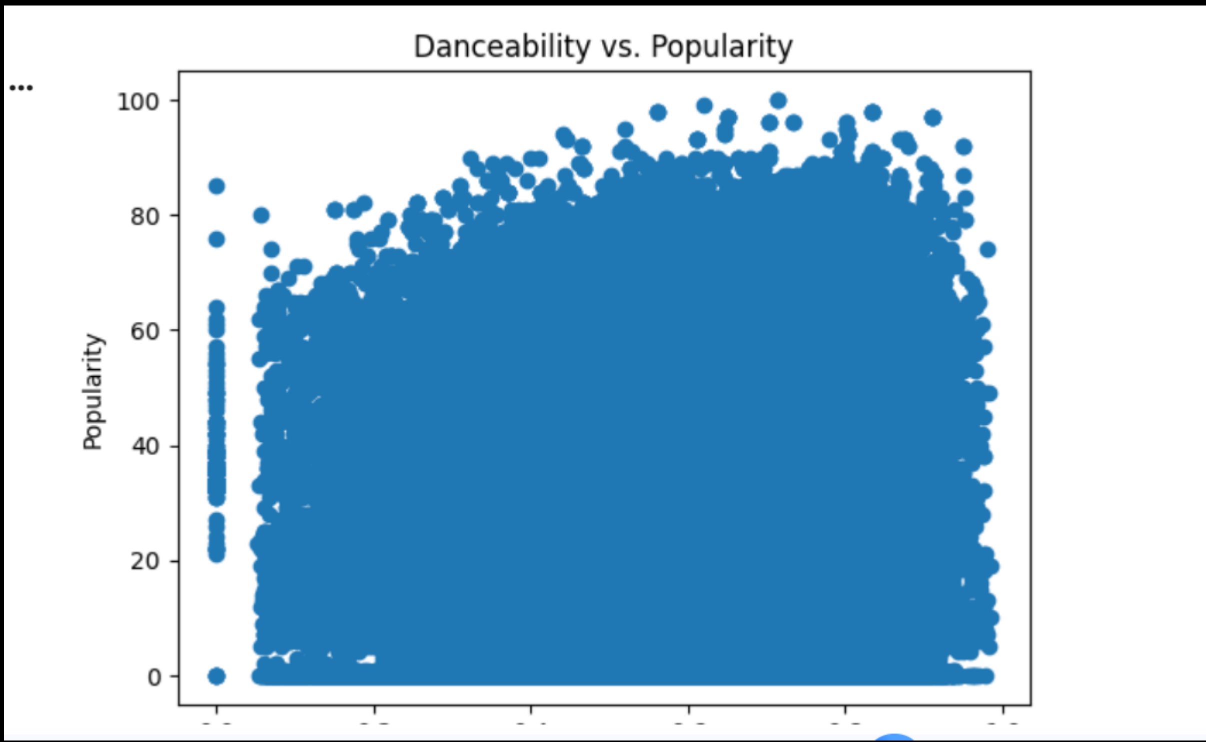

2.

```python
explicit_count = df["explicit"].value_counts()

explicit_count.plot(kind="pie", autopct="%1.1f%%")

plt.title("Explicit Songs")

plt.show()
```



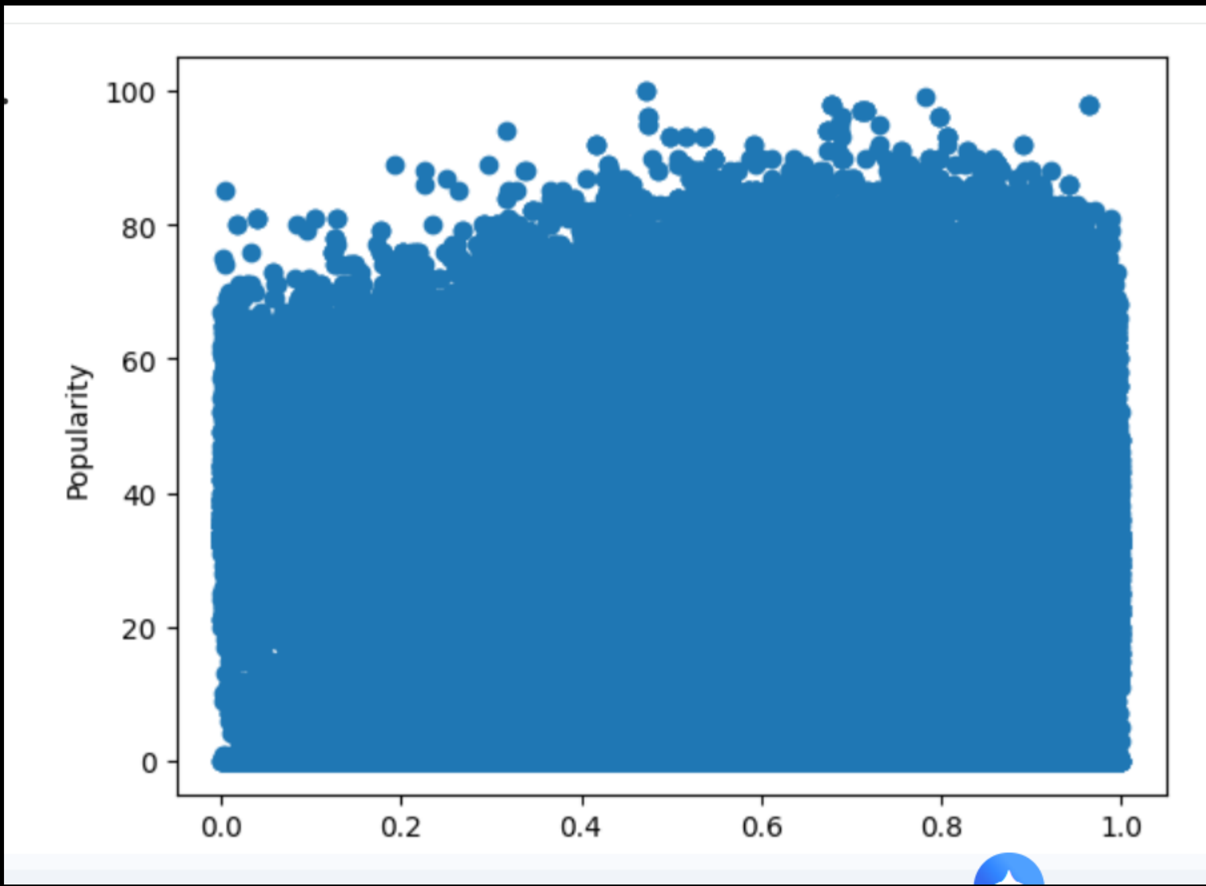

3.

```
import matplotlib.pyplot as plt

plt.scatter(df["danceability"], df["popularity"])
plt.xlabel("Danceability")
plt.ylabel("Popularity")
plt.title("Danceability vs. Popularity")
plt.show()
```




4.

```
plt.scatter(df["energy"], df["popularity"])
plt.xlabel("Energy")
plt.ylabel("Popularity")
plt.show()
```



Popularity
100
80
60
40
20
0
0.0
0.2
0.4
0.6
0.8
1.0